

MAHI AGARWAL

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Summary

Final-year Electronics and Communication Engineering undergraduate at IIIT Noida who builds full-stack applications (React, Node.js, Express, Socket.IO) as comfortably as data pipelines and analyses (SQL, Python, Excel, Power BI). Takes ambiguous, real-world problems — often straight from non-technical users — and turns them into structured data, clear business rules, and working solutions independently, end-to-end. Communicates clearly with technical and non-technical audiences. Uses AI tools regularly to move faster without cutting corners. Seeking a software or analyst role where technical ability, structured thinking, and ownership create real impact.

Technical Skills

- **Languages:** JavaScript, Python (Pandas, NumPy, scikit-learn), SQL, C++
- **Full-Stack & Backend:** React, Node.js, Express, Socket.IO (WebSockets), REST APIs, Electron
- **Data & Reporting:** SQL (MySQL, Joins, Subqueries, CTEs, Window Functions), Excel, Power BI, ETLs, MATLAB
- **Tools & Practices:** Git/GitHub, Agile/Scrum, SDLC
- **Core Strengths:** Requirements gathering from non-technical users, first-principles problem diagnosis, data-driven decision making, independent end-to-end ownership, cross-functional communication
- **Cloud & AI:** AWS, Azure (fundamentals) | GitHub Copilot, ChatGPT, Claude

Projects

OffTrack — Desktop Music Player with Spotify Sync *Electron, Node.js, yt-dlp, Spotify Web API*

- Identified a specific, unmet need — ad-free desktop listening without browser overhead — and independently designed, built, and shipped a lightweight, always-on-top Electron app end-to-end.
- Integrated the Spotify Web API for two-way playback handoff, syncing the active track and timestamp between the app and a user's Spotify session in real time.

Cartel — Real-Time Group Ordering & Bill-Splitting Platform *React, Node.js, Express, Socket.IO*

- Translated a messy real-world problem (who owes what) into clear, structured business rules: individual and opt-in shared items, with pro-rated delivery and platform fees calculated per user at checkout, in a real-time session-based platform.
- Designed a mock multi-method payment flow (UPI/Card/Cash) with live group payment-status tracking, plus an automated demo-bot system simulating concurrent users to validate real-time behaviour.

Girvi Len Den — Loan & Interest Record System (Client-Facing) *JavaScript, Structured Data Management, Excel Reporting*

- Gathered requirements directly from non-technical end users and translated their manual record-keeping process into a structured system handling per-entry rates, categories, and balances across high volumes of records.
- Implemented persistent local data storage and one-click Excel export so users could review and audit records offline without technical help.
- Owned the project end-to-end — from requirements discovery to design, build, and deployment — working directly with the end user throughout.

Edible Oil Adulteration Detection — ML Classification Pipeline *Python, scikit-learn, NumPy, Pandas*

- Built binary and multi-class Random Forest classifiers to detect and grade oil adulteration from microwave sensor (S11) frequency-response data.
- Diagnosed a data-structure bottleneck through first-principles analysis and restructured the dataset to treat each frequency-S11 reading as an individual training sample rather than one sample per file, materially improving model accuracy.

Education

Jaypee Institute of Information Technology (JIIT), Noida **July 2023 – Present**
B.Tech, Electronics and Communication Engineering

P.M.S. Public School, Moradabad **Mar 2022 – Mar 2023**
Class 12 (CBSE)

Courses & Certifications

- **Data Analytics Job Simulation** (Deloitte, via Forage; Sept 2026) — practical tasks in data analysis and forensic technology
- **AI Agents Intensive** (Google/Kaggle) — LLM-based agent workflows, prompt design, responsible AI use, GenAI prototyping